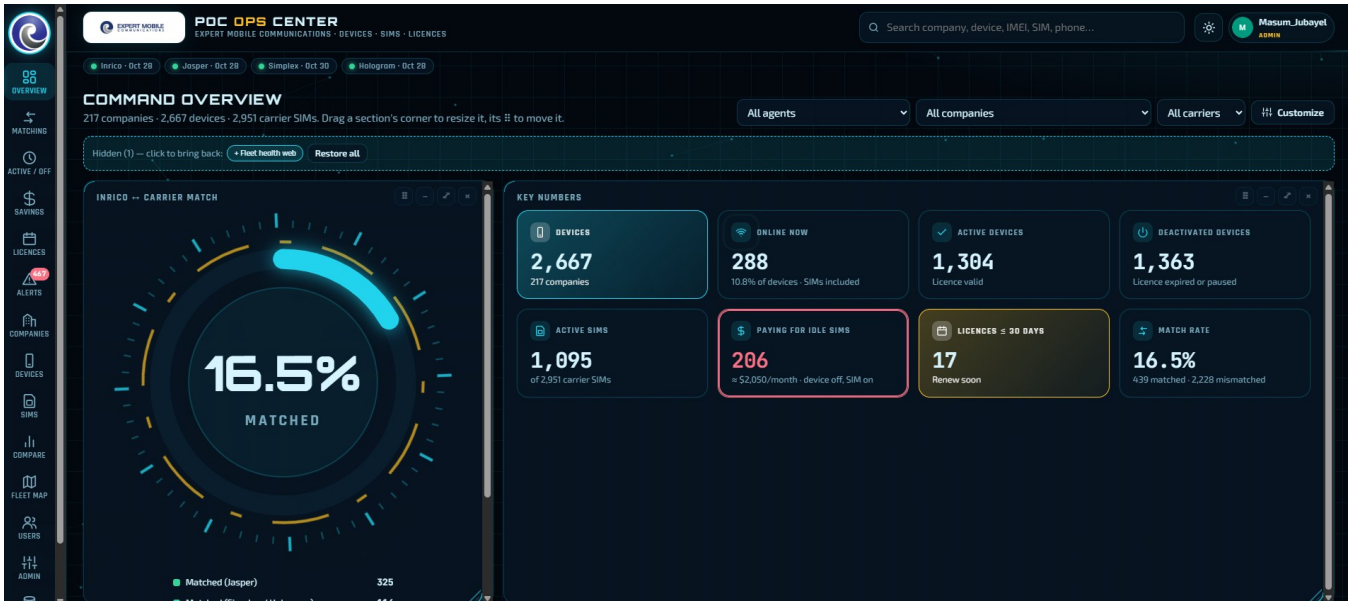
PROJECT README

POC Ops Center

A command dashboard that matches push-to-talk radios with their data SIMs across five platforms, tracks licences and puts a dollar value on idle SIMs.

Type	Internal operations platform
Organisation	Expert Mobile Communications Ltd.
Year	2026
Scale	217 companies - 2,667 devices - 2,951 carrier SIMs
Role	Designer and developer (end-to-end)
Repository	github.com/Masum-jubayel/poc-sim-operations-center
Live demo	Internal (to be added)



Command Overview: match-rate gauge, key numbers and data-source freshness. Aggregate figures only; no customer data.

Contents

1. What the project is 2. Why it was built 3. How it works 4. Features 5. Architecture 6. Technologies 7. Security & privacy 8. Results 9. Next steps

1. What the project is

POC Ops Center is the operations dashboard for Expert Mobile's push-to-talk-over-cellular (POC) radio business. It brings every radio from the Inrico management platform and every data SIM from Telus Jasper, Simplex, Hologram and Mobilewave into one live view. It matches each radio to its SIM, explains every mismatch with a reason and a severity, tracks licences before they expire, calculates what idle SIMs cost each month, and lets the team assign, follow up and export the fixes. The interface is a customizable heads-up display with a light theme and a dark Jarvis theme.

2. Why it was built

Radio licences and SIM cards lived in separate vendor portals, so simple questions had no reliable answer: which radios have no working SIM, which SIMs are billed but unused, which licences expire this month, and which account each SIM is really under. The first full import showed the size of the gap: only 16.5% of devices (439) matched cleanly, 2,228 needed attention, and 206 SIMs were active on radios that were switched off - roughly $2,050 per month of unused spend.

3. How it works, step by step

Step	Stage	What happens
1	**Import**	Each platform export (XLSX/CSV) is dropped into the Data hub; missing columns are flagged and a freshness chip records the import date.
2	**Normalise**	ICCIDs are cleaned: the trailing 'f' Inrico adds is removed, spaces and stray commas are trimmed.
3	**Match**	Radios and SIMs are joined on ICCID; customer, device name and SIM status are compared.
4	**Classify**	Each device gets a result: matched (Jasper or Simplex/Hologram), no SIM in Inrico, invalid SIM value, SIM not on any carrier, wrong Jasper customer, name differs, or same SIM on two devices.
5	**Alert**	Problems become high / medium / low alerts with a reason and a link to the details.
6	**Act**	Owners record actions (status, notes) per company; bulk actions handle common fixes.
7	**Measure**	Savings, licence expiry and match rate update as fixes land.
8	**Report**	Formatted POC and SIM workbooks, plus Excel/CSV/JSON downloads, are produced only when both platforms agree.

4. Features

- Command Overview: drag, resize, collapse, full-screen and hide widgets
- Arc-reactor style match-rate gauge with a breakdown legend
- Key numbers: devices, online now, active/deactivated devices, active SIMs, idle-SIM cost, licences due in 30 days, match rate
- Filters by agent, company and carrier applied to every widget

- Matching, Active/Off quadrant, Savings with a what-if calculator, Licences with matrix and expiry heat-map
- Alerts feed with 400+ prioritised items
- Companies, Devices and SIMs tables with health bars and a per-company drawer
- Compare view with custom groups and animated versus cards
- Fleet map treemap / heat-map
- Data hub for imports; Excel, CSV and JSON exports
- Roles: Admin, Manager, Viewer, Agent - with live presence
- Light (violet/orange) and dark Jarvis (cyan/gold) themes

5. Architecture

Five data sources feed a Data hub that validates and normalises each export. A matching engine joins and classifies the records and writes results to a shared real-time database. The dashboard, alerts, action log and report generator all read from that store, so every user sees the same picture and every change is attributed to a person.

6. Technologies used

Each technology and the job it does in this project:

Technology	What it is used for
JavaScript (ES2022)	Modern JavaScript that runs the whole dashboard in the browser: data processing, matching and interactive widgets.
HTML5 / CSS3	HTML structures each page; CSS handles layout, colour, typography and responsive design for phones and desktops.
SVG data visualisation	Gauges, donuts, treemaps, heat-maps and bar races drawn as scalable vector graphics, crisp at any size.
Real-time database	A database that pushes changes to every open screen instantly, so all users see the same live data.
Role-based access control	Each user gets a role (admin, manager, viewer, agent) that decides what they may see and change.
Python	General-purpose programming language used for data processing, automation scripts and back-end logic.
openpyxl	Python library that writes real Excel workbooks with formatting: bold headers, fills, borders, frozen panes and filters.
Excel / CSV / JSON	Export formats: Excel for people, CSV for other systems, JSON for programs and APIs.
Data matching / ETL	Extract data from several sources, Transform it to one clean format, then Load or compare it.
Browser automation	Scripts that operate web portals the way a person would (search, read tables, capture exports) when no API is available.
REST APIs	Standard web interfaces used to request and send data between the application and other services.

7. Security & privacy

- Customer names and SIM numbers are never published; only aggregate figures appear in this document.
- Role-based access: agents see only their own companies.
- Credentials stay with the signed-in user; nothing is stored in code.
- Nothing is filed while a mismatch exists.

8. Results

- One screen for 217 companies, 2,667 devices and 2,951 SIMs across five sources.
- The real starting point became visible (16.5% clean) and is worked down through tracked actions.
- 206 idle SIMs (about $2,050/month) identified for deactivation or reuse.
- Licence renewals seen 30 days ahead.
- Customer audits take minutes instead of a long session.

9. Next steps

- Scheduled imports through carrier APIs
- Weekly email summary for managers
- Trend charts for match rate and idle cost

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